

1st Sessional Test- Heat Transfer, 19.02.2026

1. A rod of 10 mm diameter and 80 mm length with thermal conductivity 16 W/m-deg protrudes from a surface at 160°C . The rod is exposed to air at 30°C with a convection coefficient of $25 \text{ W/m}^2\text{-deg}$. How does the heat flow from this rod get affected if the same material volume is used for two fins of the same length? Assume short fin with end insulated.
2. The engine oil at 150°C is cooled to 80°C in a parallel flow heat exchanger by water entering at 25°C and leaving at 60°C . Estimate the exchanger effectiveness. If the fluid flow rates and the inlet conditions remain unchanged, work out the lowest temperature to which the oil may be cooled by increasing the length of heat exchanger.

(8)

(7)